

Advanced Linear Algebra for Quantum Computing

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Topological Quantum Error Correction

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To the reader

We assume existing familiarity with sets, functions, matrices, vectors, summation notation, probabilities, and complex numbers. I recommend Dr. Beezer's free textbook [A First Course in Linear Algebra](#) to newcomers. We also assume familiarity with the mathematical basics of quantum error correction (including quantum gates and the Pauli matrices) which is given in Dr. Fowler's [free Coursera course on the subject](#).

*Narrative is linear, but action has breadth
and depth as well as height and is solid.*

—Thomas Carlyle

I discovered that the library is the real school.

—Ray Bradbury

Vector spaces

Definition

V is a vector space if it is a set of vectors coupled with the addition of vectors and scalar multiplication.

Vector addition: $\mathbf{u}, \mathbf{v} \in V$ means that $\mathbf{u} + \mathbf{v} \in V$.

Scalar multiplication: $\alpha \mathbf{v} \in V$ for any $\mathbf{v} \in V, \alpha \in \mathbb{C}$.

Remark

The vector space we often use in quantum computing is $\mathbb{C}^n$, where n is a power of 2.

Remark

Vector spaces have numerous other properties that show that vector space operations (vector addition and scalar multiplication) obey commutativity and associativity laws; they can be found [here](#).

Vector spaces (cont'd)

Definition

A subspace W of a vector space V is a subset of V and itself a vector space.

Example

For the vector space $\mathbb{C}^4$, the vector space

$$W = \left\{ \begin{pmatrix} a_0 \\ a_1 \\ a_2 \\ 0 \end{pmatrix} : a_0, a_1, a_2 \in \mathbb{C} \right\}$$

is a subspace of the vector space $\mathbb{C}^4$.

Linear independence and basis vectors

Definition

Given a set of vectors $S = \{\mathbf{v}_1, \dots, \mathbf{v}_m\}$ we say that S is linearly independent if

$$\sum_{k=1}^m \alpha_k \mathbf{v}_k = \mathbf{0},$$

where $\mathbf{0}$ is the zero vector, if $\alpha_k = 0$ for all $1 \leq k \leq m$.

Definition

If a set of vectors T can be combined in a linear fashion to produce every element of S , we say that T spans S .

Definition

Suppose V is a vector space. Then a minimum cardinality subset $B \subseteq V$ that is linearly independent and spans V is a basis of V .

Linear maps

Definition

A linear map is a function $T : U \rightarrow V$, where U and V are vector spaces, such that

$$T(\mathbf{u}_1 + \mathbf{u}_2) = T(\mathbf{u}_1) + T(\mathbf{u}_2) \text{ for all } \mathbf{u}_1, \mathbf{u}_2 \in U$$

and

$$T(\alpha \mathbf{u}) = \alpha T(\mathbf{u}) \text{ for all } \mathbf{u} \in U, \alpha \in \mathbb{C}.$$

Example

The Hadamard gate

$$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}$$

is a linear map $H : \mathbb{C}^2 \rightarrow \mathbb{C}^2$.

Dirac notation

Definition

A ket (or state vector) $|\psi\rangle$ is a normalized vector in the vector space $\mathbb{C}^n$, meaning that $\sum_{k=1}^n |\psi_k|^2 = 1$ and ψ_k is the k th element of $|\psi\rangle$.

Definition

A bra-ket $\langle\phi|\psi\rangle$ is the inner product of $|\phi\rangle$ and $|\psi\rangle$,

$$\langle\phi|\psi\rangle = \sum_{k=1}^n \phi_k^* \psi_k,$$

where the bra $\langle\phi|$ acts as a functional map (from a ket to a scalar; more on this later) on $|\psi\rangle$.

Example

For any quantum state $|\psi\rangle$, $\langle\psi|\psi\rangle = 1$, since $\psi_k \psi_k^* = |\psi_k|^2$.

Dirac notation (cont'd)

Theorem (Cauchy-Schwarz Inequality)

For any two quantum states $|\phi\rangle$ and $|\psi\rangle$, we have $|\langle\phi|\psi\rangle|^2 \leq 1$.

Definition

A ket-bra $|\phi\rangle\langle\psi|$ is a linear map (or outer product)

$$|\phi\rangle\langle\psi| = \begin{pmatrix} \phi_1 \\ \vdots \\ \phi_n \end{pmatrix} (\psi_1^*, \dots, \psi_n^*) = \begin{pmatrix} \phi_1\psi_1^* & \dots & \phi_1\psi_n^* \\ \vdots & \ddots & \vdots \\ \phi_n\psi_1^* & \dots & \phi_n\psi_n^* \end{pmatrix}$$

Remark

Given a linear map A and bra $\langle\phi|$, $\langle\phi|A$ is also a bra defined by the function composition rule

$$(\langle\phi|A)|\psi\rangle = \langle\phi|(A|\psi\rangle) = \langle\phi|A|\psi\rangle$$

Orthonormal basis

Definition

An orthonormal basis is a set of basis vectors $\mathcal{B}$ in a vector space with a Euclidean norm of 1, meaning $\sqrt{\sum_{k=1}^n |\psi_k|^2} = 1$ for all $|\psi\rangle \in \mathcal{B}$, and every pair of distinct vectors $|\phi\rangle, |\psi\rangle \in \mathcal{B}$ is orthogonal, meaning $\langle\phi|\psi\rangle = 0$.

Example

The Euclidean basis vectors form an orthonormal basis of $\mathbb{C}^n$; for $n = 4$ they are

$$\left\{ \begin{pmatrix} 1 \\ 0 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 0 \\ 1 \end{pmatrix} \right\}$$

Linear functionals and dual spaces

Definition

A linear functional f is a mapping between elements of a vector space into the complex plane (ie, $f : V \rightarrow \mathbb{C}$) such that

$$f(\mathbf{u} + \mathbf{v}) = f(\mathbf{u}) + f(\mathbf{v}) \text{ and } f(\alpha\mathbf{u}) = \alpha f(\mathbf{u})$$

for any $\mathbf{u}, \mathbf{v} \in V$ and $\alpha \in \mathbb{C}$.

Definition

The dual space of a vector field V is the set of all linear functionals over V .

Remark

The bra $\langle\phi|$ is a member of the dual space of the vector space containing $|\phi\rangle$. As stated previously, the bra-ket represents the mapping of state vectors in $\mathbb{C}^n$ into $\mathbb{C}$.

Eigenstates

Definition

A state vector $|\psi\rangle$ is the λ -eigenstate (or eigenvector) of a linear map U if $U|\psi\rangle = \lambda|\psi\rangle$ for some $\lambda \in \mathbb{R}$, where λ is said to be an eigenvalue of U (and corresponds to a measurement of a qubit).

Example

If $U|\psi\rangle = |\psi\rangle$, we say that $|\psi\rangle$ is the $+1$ eigenstate of U .

Exercise

Show that $|+\rangle$ is the $+1$ eigenstate of $X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Exercise

Show that $|0\rangle$ and $|1\rangle$ are the $+1$ and -1 eigenstates of $Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

Unitary matrices

Definition

The conjugate of a matrix U is denoted U^* and obtained by conjugating each of its elements.

Definition

The transpose of a matrix U is denoted U^T and is obtained by systematically exchanging (i.e., swapping) the values of the rows with the values in the columns in U .

Definition

The adjoint of a matrix U is the conjugate transpose

$$U^\dagger = (U^*)^T = \begin{pmatrix} u_{1,1}^* & \cdots & u_{1,n}^* \\ \vdots & \ddots & \vdots \\ u_{n,1}^* & \cdots & u_{n,n}^* \end{pmatrix}^T = \begin{pmatrix} u_{1,1}^* & \cdots & u_{n,1}^* \\ \vdots & \ddots & \vdots \\ u_{1,n}^* & \cdots & u_{n,n}^* \end{pmatrix}$$

Unitary matrices (cont'd)

Definition

A matrix U is Hermitian if it is self-adjoint, that is, $U = U^\dagger$.

Definition

A matrix U is unitary if $UU^\dagger = U^\dagger U = I$.

Exercise

Show that

$$U = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$$

is unitary but not Hermitian.

Exercise

Show that every quantum operator is unitary.

Hint. Use the fact that $\langle \psi | \psi \rangle = 1$ and substitute $|\psi\rangle$ for $U|\psi\rangle$ for an arbitrary operator U . Then use function composition rules.

The Kronecker product

Definition

For two-dimensional, normalized quantum states $|\phi\rangle$ and $|\psi\rangle$, the Kronecker product (or tensor product) between them is

$$|\phi\rangle \otimes |\psi\rangle = \begin{pmatrix} \phi_1 \\ \phi_2 \end{pmatrix} \otimes \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} = \begin{pmatrix} \phi_1 \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} \\ \phi_2 \begin{pmatrix} \psi_1 \\ \psi_2 \end{pmatrix} \end{pmatrix} = \begin{pmatrix} \phi_1\psi_1 \\ \phi_1\psi_2 \\ \phi_2\psi_1 \\ \phi_2\psi_2 \end{pmatrix}$$

and is sometimes denoted $|\phi\rangle |\psi\rangle$ or $|\phi\psi\rangle$.

Remark

You can take Kronecker products of matrices as well; the new object's dimension is the product of the dimensions of the objects multiplied, meaning that Kronecker products of higher-dimensional objects quickly become unmanageable to compute by hand.

The Kronecker product (cont'd)

Definition

The Kronecker product of two 2×2 matrices A and B can be computed as

$$A \otimes B = \begin{pmatrix} a_{11}B & a_{12}B \\ a_{21}B & a_{22}B \end{pmatrix},$$

which is a 4×4 matrix.

Exercise

Compute the Kronecker products $X \otimes X$ and $T \otimes T$, where

$$T = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/4} \end{pmatrix}.$$

God created the integers; all else is the work of man.
—Leopold Kronecker

The commutator

Definition

Two matrices A and B are said to commute if $AB = BA$.

Remark

For quantum operators A and B , $(AB)|\psi\rangle$ means “apply B first, then A , on $|\psi\rangle$.”

Definition

The commutator of two matrices is defined as $[A, B] = AB - BA$.

Corollary

For commuting matrices, $[A, B] = \mathcal{O}$ (the zero matrix).

Exercise

Show that $[X, Z] = -2iY$, where $Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$.

Matrix trace

Definition

The trace of an n -dimensional matrix A is the sum of its diagonal elements; that is

$$\operatorname{tr}(A) = \sum_{k=1}^n A_{k,k}$$

Remark

Trace is “commutativity-preserving,” meaning $\operatorname{tr}(AB) = \operatorname{tr}(BA)$ for any matrices A, B .

Remark

The trace is independent of a chosen basis, meaning that if a linear map A , represented by a matrix, has a “change-of-basis” matrix P such that $B = P^{-1}AP$, then $\operatorname{tr}(A) = \operatorname{tr}(B)$.

Riesz Representation Theorem

Definition

A Hilbert space $\mathcal{H}$ in quantum computing is a complex inner product space, that is, a vector space over $\mathbb{C}^n$ endowed with an inner product operation.

Definition

A linear map T is anti-linear if

$$T(\alpha |\phi\rangle + \beta |\psi\rangle) = \alpha^* T |\phi\rangle + \beta^* T |\psi\rangle .$$

Definition

Two vector spaces V and W are isomorphic if there exists a bijective linear map $T : V \rightarrow W$, meaning that

- ▶ (it's injective) every unique vector in V maps to a unique vector in W (that is, if $T(\mathbf{u}) = T(\mathbf{v})$ then $\mathbf{u} = \mathbf{v}$) and
- ▶ (it's surjective) that for every vector $\mathbf{w} \in W$ there exists a vector $\mathbf{v} \in V$ such that $T(\mathbf{v}) = \mathbf{w}$.

Riesz Representation Theorem (cont'd)

Definition

An isomorphism is canonical if it is defined by the intrinsic mathematical properties of the vector spaces it acts on, that is, no additional change-of-basis matrix is necessary to perform it.

Theorem (Riesz)

For any Hilbert space $\mathcal{H}$ there exists a unique canonical anti-linear isomorphism between $\mathcal{H}$ and its dual space $\mathcal{H}^*$.

Remark

The Riesz representation theorem is how we ensure a bijective, normalization-preserving correspondence between bras and kets; we will often write $|\psi\rangle = (\langle\psi|)^\dagger$, where the Hermitian conjugate operator $\dagger$ represents the conversion between a row and a column vector and conjugation of vector elements.

Projection operators

Definition

A projection operator, denoted $\hat{P}$, when applied to a quantum state $|\psi\rangle \in \mathcal{H}$, where $\mathcal{H}$ is a Hilbert space, returns a vector in a subspace of $\mathcal{H}$ defined with respect to a specific state $|a\rangle$ as

$$\hat{P}_a = |a\rangle \langle a|,$$

where $|a\rangle$ is in the orthonormal basis of $\mathcal{H}$.

Remark

Projection operators are used to model quantum measurement; in quantum computing, they project a qubit into a specific basis, such as the X or Z basis (where only one of two outcomes are possible).

Remark

Projection operators are idempotent, meaning that $\hat{P}^2 = \hat{P}$, and Hermitian, meaning that $\hat{P}^\dagger = \hat{P}$.

Projection operators (cont'd)

Example

For a single qubit in the general state $|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$, a measurement in the computational basis $Z = \{|0\rangle, |1\rangle\}$ is described by two projection operators:

- ▶ Projection operator for outcome $|0\rangle$: $\hat{P}_0 = |0\rangle\langle 0|$; note that $p(0) = |\alpha|^2$
- ▶ Projection operator for outcome $|1\rangle$: $\hat{P}_1 = |1\rangle\langle 1|$; note that $p(1) = |\beta|^2$

Exercise

Compute the matrices $\hat{P}_0$ and $\hat{P}_1$.

Remark

The probability of measuring $|\psi\rangle$ as $|a\rangle$ can be computed as

$$p(a) = |\langle a|\psi\rangle|^2 = \langle\psi|\hat{P}_a|\psi\rangle.$$

Spectral Theorem

Definition

A diagonal matrix is a matrix such that all entries outside the main diagonal are zero.

Definition

The spectrum of a matrix A is the set of all eigenvalues of A .

Definition

A matrix A is normal if $[A, A^\dagger] = \mathcal{O}$.

Theorem

If a matrix A is normal, then A is unitarily diagonalizable, meaning that $A = UDU^\dagger$ for some unitary matrix U comprising the eigenvectors of A , and a diagonal matrix D , where the diagonal values of D is the spectrum of A . Such a factorization is called the spectral decomposition of A .

Spectral Theorem (cont'd)

Remark

The spectral decomposition of any n -dimensional Hermitian matrix A is

$$A = \sum_{k=1}^n \lambda_k |\psi\rangle_k \langle\psi|_k,$$

where λ_k is an eigenvalue of A and $|\psi\rangle_k$ is its corresponding eigenstate.

Example

The spectral theorem yields the following decomposition of the Pauli Z matrix in terms of projection operators:

$$Z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = (1)\hat{P}_1 + (-1)\hat{P}_{-1}$$

since the eigenvalues of Z are $+1$ and -1 .

Spectral Theorem (cont'd)

Remark

The spectral theorem tells us that, since the Pauli matrices $\{X, Y, Z\}$ are normal, their eigenvalues are ± 1 and their eigenvectors form a mutually orthogonal and complete basis of the qubit state space $\mathbb{C}^2$, justifying their use as the fundamental observables of a qubit, since the Pauli matrices create a basis for all 2×2 Hermitian matrices, which

Exercise

Show that every unitary matrix and Hermitian matrix is normal.

This isn't right. This isn't even wrong.
—Wolfgang Pauli